

ECONOMICS

Week 01

Introduction - Basics of Economics

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Course Material

- Lectures
 - Class Notes
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- **Recommended Books**

- William G. Sullivan, James A. Bontadelli, Elin M. Wicks, “Engineering Economy” (14th Edition).
 - Karl E. Case, Ray C. Fair, Sharon M. Oster, “Principles of Economics” (10th Edition).
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Note:

Books and Lectures will be available in the Google Classroom.

Grading Policy

No.	Type of Work	Weightage
1	Quizzes	10%
2	Assignments	10%
3	Project	10%
4	Sessional	30%
5	Final Exam	40%

Program Learning Outcomes

- PLO 2. **Problem Analysis**: An ability to identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
- PLO 12. **Lifelong learning**: An ability to recognize importance of, and pursue lifelong learning in the broader context of innovation and technological developments.

Course Learning Outcomes

- CLO 1. Define and understand the different economic principles applied in business and manufacturing and service concerns. [Cognitive, C1]
- CLO 2. Apply the cost analysis techniques to compute demand, supply, breakeven analysis of an engineering work or an ongoing project. [Cognitive, C3]

Adam Smith (1723 – 1790)

- Known as father of modern economics
- Wrote the first and most important books on the subject of economics,
 - The Theory of Moral Sentiments (1759)
 - An Inquiry into the Nature and Causes of the Wealth of Nations (1776)
- Smith believed in “free market”
 - (free market is one where consumers may buy what they like and producers may produce what they like, with no government interference)
- Smith suggested that a laissez-faire (don't interfere) approach should be followed, leaving customers and producers to make their own decisions



Economics

- Economics is a social science that analyzes the production, distribution and consumption of goods and services.
- It is the study of how individuals and societies choose to use the scarce resources that exist in nature to fulfill their wants and needs through a decision-making mechanism, social customs, and political realities of the society.
- **The three central coordination problems any economy must solve:**
 1. What, and how much to produce
 2. How to produce it
 3. For whom to produce it

Objectives

The objective is the goal or desired outcome of a choice

- Individuals may try to maximize utility given the constraints of income, time, prices, etc.
- Firms may have objectives such as the maximization of profits, sales, market share, etc. or the minimization of costs per unit
- Social objective, maximize the well being of the members of society

Microeconomics and Macroeconomics

- **Microeconomics** studies such things as:
 - The pricing policy of firms
 - Household's decisions on what to buy
 - How markets allocate resources among alternative ends
- **Macroeconomics** studies such things as:
 - Inflation
 - Unemployment
 - Economic growth

The Scope of Economics

Examples of microeconomic and macroeconomic concerns

	Production	Prices	Income	Employment
Microeconomics	Production/Output in Individual Industries and Businesses How much steel How many offices How many cars	Price of Individual Goods and Services Price of medical care Price of gasoline Food prices Apartment rents	Distribution of Income and Wealth Wages in the auto industry Minimum wages Executive salaries Poverty	Employment by Individual Businesses & Industries Jobs in the steel industry Number of employees in a firm
Macroeconomics	National Production/Output Total Industrial Output Gross Domestic Product Growth of Output	Aggregate Price Level Consumer prices Producer Prices Rate of Inflation	National Income Total wages and salaries Total corporate profits	Employment and Unemployment in the Economy Total number of jobs Unemployment rate

Economic Institutions

- To apply economic theory to reality, you've got to have a sense of economic institutions
- **Economic institutions** are laws, common practices, and organizations in a society that affect the economy
- Economic institutions differ significantly among nations
- They sometimes seem to operate differently than economic theory predicts

Economic Policy Options

- **Economic policies** are actions (or inactions) taken by the government to influence economic actions
- **Objective** policy analysis keeps value judgments separate from the analysis
- **Subjective** policy analysis reflects the analyst's views of how things should be

Scarcity

- Scarcity exists because individuals want more than can be produced
- **Scarcity** means the goods available are too few to satisfy individuals' desires
- The degree of scarcity is constantly changing
- The quantity of goods, services and usable resources depends on technology and human action

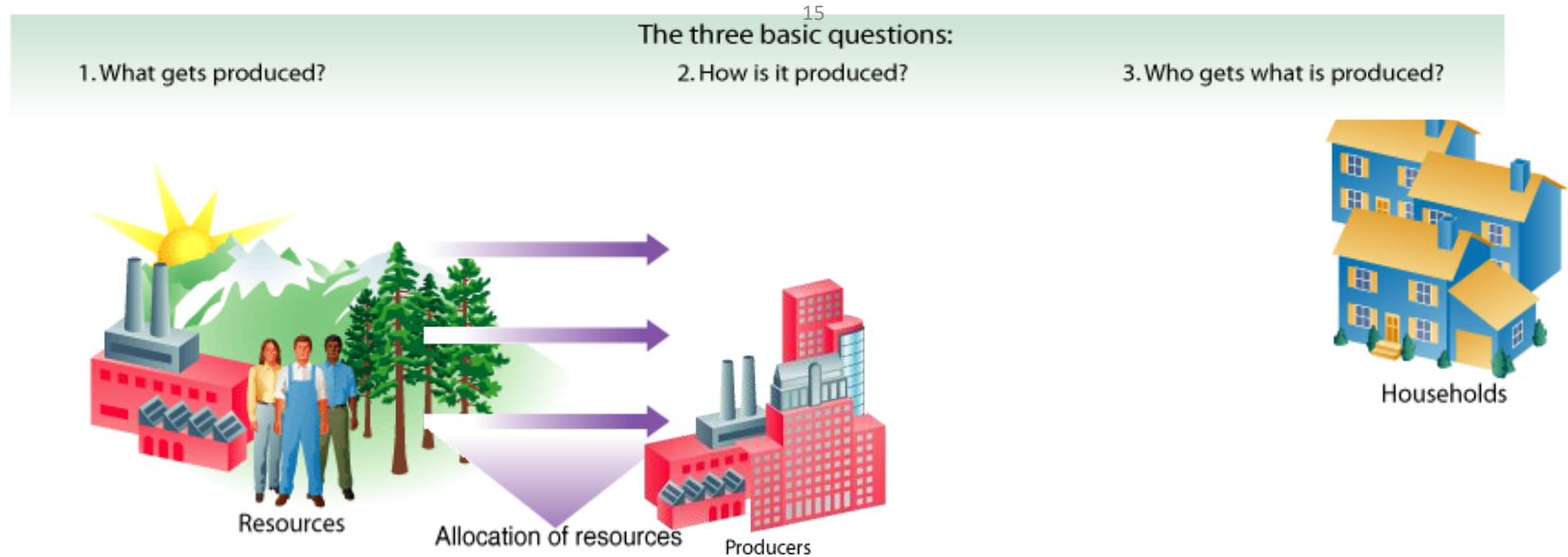
The Economic Problem: Scarcity & Choice

Economics is the study of choices when there is scarcity.

Here are some examples of scarcity and the trade-offs associated with making choices:

- You have a limited amount of time. If you take a part-time job, each hour on the job means one less hour for study or play.
- A city has a limited amount of land. If the city uses an acre of land for a park, it has one less acre for housing, retailers, or industry.
- You have limited income this year. If you spend \$17 on a music CD, that's \$17 less you have to spend on other products or to save.

The Economic Problem: Scarcity & Choice



Three basic questions must be answered in order to understand an economic system:

What gets produced?

How is it produced?

Who gets what is produced?

Factors of Production

The resources used to produce goods and services; also known as production inputs.

- **Natural resources/Land**

Resources provided by nature and used to produce goods and services

- **Labor**

The physical and mental effort people use to produce goods and services

- **Human capital**

The knowledge and skills acquired by a worker through education and experience.

- **Physical capital & Financial Capital**

The stock of equipment, machines, structures, and infrastructure that is used to produce goods and services.

Input of funds

- **Entrepreneurship**

The effort used to coordinate the factors of production—natural resources, labor, physical capital, and human capital—to produce and sell products.

Economic Terminology

- Goal of this class is to describe how economics works in the real world
- You will be introduced to many terms that occur in business and in discussions of the economy
- **For example:**
 - Opportunity cost
 - Marginal benefit and marginal cost
 - Market and economic forces
 - Demand Supply
 - Production Function
 - Optimization, etc.

Opportunity Cost

- **Opportunity cost** is the benefit forgone of the next-best alternative to the activity you have chosen
- Opportunity cost should always be less than the benefit of what you have chosen
- Opportunity cost is the basis of cost/benefit economic reasoning

Marginal Costs and Marginal Benefits

- Using economic reasoning, decisions are often made by comparing marginal costs and marginal benefits
- **Marginal cost** is the additional cost over and above costs already incurred
- **Marginal benefit** is the additional benefit above and beyond what has already accrued

Marginal Costs and Marginal Benefits

The economic decision rule:

- If the marginal benefits of doing something exceed the marginal costs, do it.

$MB > MC \rightarrow \text{Do it!}$

- If the marginal costs of doing something exceed the marginal benefits, don't do it.

$MC > MB \rightarrow \text{Don't do it!}$

Opportunity Cost

Examples of opportunity cost:

1. Individual decisions

The opportunity cost of college includes:

- Items you could have purchased with the money spent for tuition and books
- Loss of the income from a full-time job

2. Government decisions

- The opportunity cost of money spent on the war on terrorism is less spending on health care or education

Economic and Market Forces

- **Economic forces** are mechanisms that ration scarce goods
- A **market force** is an economic force that is given relatively free rein by society to work through the market
- **The invisible hand** is the price mechanism that guides our actions in a market. The invisible hand is an example of a market force.
- If there is a *shortage*, prices rise
- If there is a *surplus*, prices fall

Economic and Market Forces

- What happens in society can be seen as a reaction to, and interaction of:
 - Economic forces
 - Social forces
 - Historical forces
- Social, cultural, and political forces influence market forces
- Political and social forces often work together against the invisible hand

Economic Insights

- Theories tie together economists' terminology and knowledge about economic institutions
- Theories are too abstract to apply in specific cases and are often embodied in economic models and principles
- An **economic model** is a framework that places the generalized insights of the theory in a more specific contextual setting
- An **economic principle** is a commonly held insight stated as a law or general assumption

The Invisible Hand Theory

- According to the invisible hand theory, a market economy, through the price mechanism, will allocate resources efficiently
 - Prices *fall* when quantity supplied is greater than quantity demanded
 - Prices *rise* when the quantity demanded is greater than the quantity supplied
- **Efficiency** means achieving a goal as cheaply as possible